

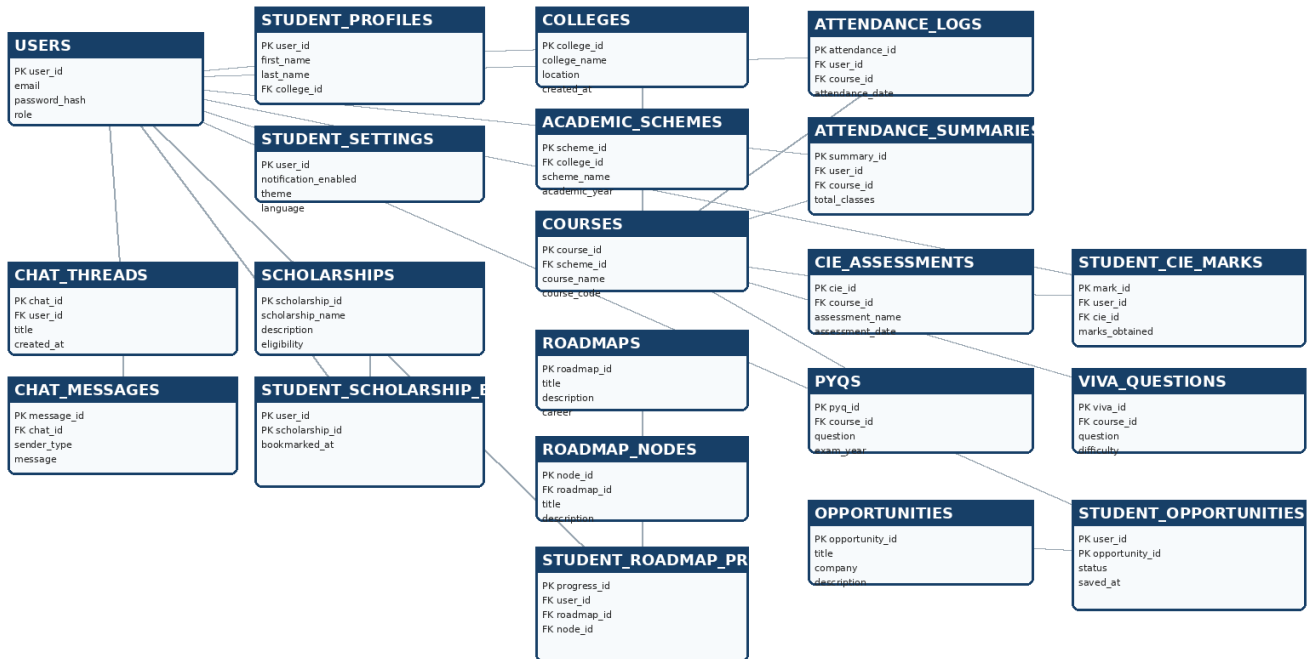
CampusOS – Corrected Database Schema

Normalized ER design, entities, keys, relationships, validation rules, and implementation guidance

Purpose: This document converts the uploaded CampusOS ER diagram into a consistent relational database design. It corrects duplicate entities, PK/FK mismatches, unclear columns, incorrect cardinalities, and normalization issues.

1. Final ER Diagram

The corrected logical structure uses **USERS** as the central identity entity and separates master data from student-specific, transaction, bookmark, and progress tables.



2. Design Principles

- **Consistent identifiers:** IDs use the same name and datatype wherever referenced.
- **Primary keys:** Every entity has a clear PK; bridge tables use composite PKs where appropriate.
- **Foreign-key integrity:** Student-specific records reference USERS.user_id.
- **Normalization:** Master data is separated from transactions, bookmarks, and progress.
- **Correct cardinality:** One-to-many relationships place the FK on the many side.
- **No duplicate entities:** COLLEGES, CHAT_THREADS, and VIVA_QUESTIONS each appear only once.
- **Scalable AI chat:** One thread contains many messages.
- **Scalable roadmap:** ROADMAPS, ROADMAP_NODES, and student progress are separated.
- **Many-to-many modeling:** Scholarship bookmarks and opportunity tracking use bridge tables.

3. Complete Table Definitions

1. USERS

Central identity table.

Column	Data Type	Key / Constraint
user_id	BIGINT	PK
email	VARCHAR(255)	UNIQUE, NOT NULL
password_hash	TEXT	NOT NULL
role	VARCHAR(30)	student/admin/faculty
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

2. STUDENT_PROFILES

One-to-one student profile extension of USERS.

Column	Data Type	Key / Constraint
user_id	BIGINT	PK, FK → USERS.user_id
first_name	VARCHAR(100)	NOT NULL
last_name	VARCHAR(100)	NULL
college_id	BIGINT	FK → COLLEGES.college_id
course_id	BIGINT	FK → COURSES.course_id
semester	SMALLINT	NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

3. STUDENT_SETTINGS

One settings record per user.

Column	Data Type	Key / Constraint
user_id	BIGINT	PK, FK → USERS.user_id
notification_enabled	BOOLEAN	DEFAULT TRUE
theme	VARCHAR(20)	DEFAULT 'system'
language	VARCHAR(20)	DEFAULT 'en'

4. COLLEGES

Single canonical college table.

Column	Data Type	Key / Constraint
college_id	BIGINT	PK
college_name	VARCHAR(255)	NOT NULL
location	VARCHAR(255)	NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

5. ACADEMIC_SCHEMES

Academic scheme belonging to a college.

Column	Data Type	Key / Constraint
scheme_id	BIGINT	PK
college_id	BIGINT	FK → COLLEGES.college_id
scheme_name	VARCHAR(100)	NOT NULL
academic_year	VARCHAR(20)	NULL

6. COURSES

Course belonging to an academic scheme.

Column	Data Type	Key / Constraint
course_id	BIGINT	PK
scheme_id	BIGINT	FK → ACADEMIC_SCHEMES.scheme_id
course_name	VARCHAR(255)	NOT NULL
course_code	VARCHAR(50)	UNIQUE

7. ATTENDANCE_LOGS

Individual attendance event.

Column	Data Type	Key / Constraint
attendance_id	BIGINT	PK
user_id	BIGINT	FK → USERS.user_id
course_id	BIGINT	FK → COURSES.course_id
attendance_date	DATE	NOT NULL
status	VARCHAR(20)	present/absent/late

8. ATTENDANCE_SUMMARIES

Course-wise attendance summary.

Column	Data Type	Key / Constraint
summary_id	BIGINT	PK
user_id	BIGINT	FK → USERS.user_id
course_id	BIGINT	FK → COURSES.course_id
total_classes	INT	NOT NULL
attended_classes	INT	NOT NULL
attendance_percentage	DECIMAL(5,2)	NOT NULL

9. CIE_ASSESSMENTS

Defines each CIE assessment.

Column	Data Type	Key / Constraint
cie_id	BIGINT	PK

Column	Data Type	Key / Constraint
course_id	BIGINT	FK → COURSES.course_id
assessment_name	VARCHAR(100)	NOT NULL
assessment_date	DATE	NULL
max_marks	DECIMAL(6,2)	NOT NULL

10. STUDENT_CIE_MARKS

Student marks for each CIE assessment.

Column	Data Type	Key / Constraint
mark_id	BIGINT	PK
user_id	BIGINT	FK → USERS.user_id
cie_id	BIGINT	FK → CIE_ASSESSMENTS.cie_id
marks_obtained	DECIMAL(6,2)	NOT NULL

11. PYQS

Previous-year questions linked to courses.

Column	Data Type	Key / Constraint
pyq_id	BIGINT	PK
course_id	BIGINT	FK → COURSES.course_id
question	TEXT	NOT NULL
exam_year	SMALLINT	NULL
marks	DECIMAL(5,2)	NULL
difficulty	VARCHAR(20)	easy/medium/hard

12. VIVA_QUESTIONS

Single canonical viva question table.

Column	Data Type	Key / Constraint
viva_id	BIGINT	PK
course_id	BIGINT	FK → COURSES.course_id
question	TEXT	NOT NULL
difficulty	VARCHAR(20)	easy/medium/hard

13. CHAT_THREADS

One user can have many AI Mentor chats.

Column	Data Type	Key / Constraint
chat_id	BIGINT	PK
user_id	BIGINT	FK → USERS.user_id
title	VARCHAR(255)	NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

14. CHAT_MESSAGES

Many messages belong to one chat.

Column	Data Type	Key / Constraint
message_id	BIGINT	PK
chat_id	BIGINT	FK → CHAT_THREADS.chat_id

Column	Data Type	Key / Constraint
sender_type	VARCHAR(20)	user/assistant
message	TEXT	NOT NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

15. SCHOLARSHIPS

Master list of scholarships.

Column	Data Type	Key / Constraint
scholarship_id	BIGINT	PK
scholarship_name	VARCHAR(255)	NOT NULL
description	TEXT	NULL
eligibility	TEXT	NULL
application_url	TEXT	NULL
deadline	DATE	NULL

16. STUDENT_SCHOLARSHIP_BOOKMARKS

Many-to-many scholarship bookmark bridge.

Column	Data Type	Key / Constraint
user_id	BIGINT	PK, FK → USERS.user_id
scholarship_id	BIGINT	PK, FK → SCHOLARSHIPS.scholarship_id
bookmarked_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

17. ROADMAPS

Master career roadmap.

Column	Data Type	Key / Constraint
roadmap_id	BIGINT	PK
title	VARCHAR(255)	NOT NULL
description	TEXT	NULL
career	VARCHAR(150)	NULL

18. ROADMAP_NODES

Ordered nodes within a roadmap.

Column	Data Type	Key / Constraint
node_id	BIGINT	PK
roadmap_id	BIGINT	FK → ROADMAPS.roadmap_id
title	VARCHAR(255)	NOT NULL
description	TEXT	NULL
sequence_no	INT	NOT NULL

19. STUDENT_ROADMAP_PROGRESS

Student progress for roadmap nodes.

Column	Data Type	Key / Constraint
progress_id	BIGINT	PK
user_id	BIGINT	FK → USERS.user_id
roadmap_id	BIGINT	FK → ROADMAPS.roadmap_id

Column	Data Type	Key / Constraint
node_id	BIGINT	FK → ROADMAP_NODES.node_id
status	VARCHAR(20)	not_started/in_progress/completed
completed_at	TIMESTAMP	NULL

20. OPPORTUNITIES

Master list of jobs, internships, hackathons, etc.

Column	Data Type	Key / Constraint
opportunity_id	BIGINT	PK
title	VARCHAR(255)	NOT NULL
company	VARCHAR(255)	NULL
description	TEXT	NULL
application_url	TEXT	NULL
deadline	DATE	NULL

21. STUDENT_OPPORTUNITIES

Many-to-many user/opportunity tracking bridge.

Column	Data Type	Key / Constraint
user_id	BIGINT	PK, FK → USERS.user_id
opportunity_id	BIGINT	PK, FK → OPPORTUNITIES.opportunity_id
status	VARCHAR(30)	saved/applied/shortlisted/rejected
saved_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

4. Relationship & Cardinality Reference

Relationship	Cardinality	Meaning
USERS → STUDENT_PROFILES	1 : 0..1	One user may have one student profile.
USERS → STUDENT_SETTINGS	1 : 0..1	One user may have one settings record.
COLLEGES → STUDENT_PROFILES	1 : N	One college can have many students.
COLLEGES → ACADEMIC_SCHEMES	1 : N	One college can offer many schemes.
ACADEMIC_SCHEMES → COURSES	1 : N	One scheme can contain many courses.
USERS → ATTENDANCE_LOGS	1 : N	A student has many attendance records.
COURSES → ATTENDANCE_LOGS	1 : N	A course has many attendance records.
USERS → ATTENDANCE_SUMMARIES	1 : N	A student can have summaries for many courses.
COURSES → ATTENDANCE_SUMMARIES	1 : N	A course has summaries for many students.
COURSES → CIE_ASSESSMENTS	1 : N	A course can have multiple assessments.
USERS → STUDENT_CIE_MARKS	1 : N	A student can have many marks.
CIE_ASSESSMENTS → STUDENT_CIE_MARKS	1 : N	One assessment has marks for many students.
COURSES → PYQS	1 : N	A course can have many PYQs.
COURSES → VIVA_QUESTIONS	1 : N	A course can have many viva questions.
USERS → CHAT_THREADS	1 : N	A user can create multiple chat threads.
CHAT_THREADS → CHAT_MESSAGES	1 : N	A chat contains multiple messages.
USERS ↔ SCHOLARSHIPS	N : M	Implemented by STUDENT_SCHOLARSHIP_BOOKMARKS.
ROADMAPS → ROADMAP_NODES	1 : N	A roadmap contains ordered nodes.
USERS → STUDENT_ROADMAP_PROGRESS	1 : N	A student can have many progress records.
ROADMAP_NODES → STUDENT_ROADMAP_PROGRESS	1 : N	A node can have progress records for many students.
USERS ↔ OPPORTUNITIES	N : M	Implemented by STUDENT_OPPORTUNITIES.

5. Important Corrections from the Original Diagram

Original issue	Corrected design
Duplicate COLLEGES tables	Merged into one canonical COLLEGES table with college_id as PK.
Duplicate VIVA_QUESTIONS	Merged into one VIVA_QUESTIONS table.
Duplicate CHAT_THREADS	Merged into one CHAT_THREADS table.
USERS / STUDENT_PROFILES datatype mismatch	user_id is BIGINT in both tables and the profile PK is also an FK.
Unclear course identifiers	Standardized to course_id.
CHAT_MESSAGES using chat_id as PK	Changed to message_id PK + chat_id FK, allowing multiple messages per chat.
STUDENT_CIE_MARKS using only user_id as PK	Changed to mark_id PK plus user_id and cie_id FKs.
Scholarship bookmarks using student_id alone	Changed to composite PK (user_id, scholarship_id).
ROADMAPS mixing roadmap and node data	Separated into ROADMAPS and ROADMAP_NODES.
ROADMAP progress using user_id alone	Changed to progress_id and added roadmap_id/node_id references.
OPPORTUNITIES mixing master and user data	Separated master OPPORTUNITIES from STUDENT_OPPORTUNITIES.
Ambiguous attendance fields	Added user_id, course_id, attendance_date, and status.

6. Recommended Constraints

- USERS.email should be UNIQUE and NOT NULL.
- COURSES.course_code should be UNIQUE when a course code exists.

- STUDENT_CIE_MARKS should have UNIQUE(user_id, cie_id) if a student receives only one mark per assessment.
- STUDENT_SCHOLARSHIP_BOOKMARKS uses (user_id, scholarship_id) as its composite primary key.
- STUDENT_OPPORTUNITIES uses (user_id, opportunity_id) as its composite primary key.
- ROADMAP_NODES should have UNIQUE(roadmap_id, sequence_no).
- Attendance percentage should normally be derived from attended_classes / total_classes.
- Use CHECK constraints for status values such as attendance and roadmap progress.

7. PostgreSQL Implementation Order

- USERS
- COLLEGES
- ACADEMIC_SCHEMES
- COURSES
- STUDENT_PROFILES
- STUDENT_SETTINGS
- ATTENDANCE_LOGS
- ATTENDANCE_SUMMARIES
- CIE_ASSESSMENTS
- STUDENT_CIE_MARKS
- PYQS
- VIVA_QUESTIONS
- CHAT_THREADS
- CHAT_MESSAGES
- SCHOLARSHIPS
- STUDENT_SCHOLARSHIP_BOOKMARKS
- ROADMAPS
- ROADMAP_NODES
- STUDENT_ROADMAP_PROGRESS
- OPPORTUNITIES
- STUDENT_OPPORTUNITIES

8. Final Validation Checklist

- Every table has a clear primary key.
- Every FK references a matching PK datatype.
- No duplicate table names remain.
- One-to-many relationships place the FK in the child table.
- Many-to-many relationships use bridge tables.
- Chat threads support multiple messages.
- CIE assessments support many students and multiple assessments.
- Scholarship bookmarks support many scholarships per student.
- Roadmaps support multiple nodes and student-level progress.
- Opportunities can exist independently and can then be saved/applied to by users.
- The schema is suitable as the logical foundation for PostgreSQL.

Final recommendation: Use this corrected schema as the baseline for CampusOS implementation. Before production deployment, finalize business rules for roles, semester structure, CIE retakes, attendance statuses, roadmap completion, and opportunity application states.